

Table 1: Available future-state results on 297 held-out pairs (94 patients). All scores are multiplied by 100 and shown to two decimals; arrows indicate the preferred direction. ZS denotes public-checkpoint zero-shot inference.

Method	Future clinical status		Disease progression		Future report fidelity		Probabilistic reliability	
	AP ↑	AUROC ↑	Trans. F1↑	Dir. F1↑	RadGraph F1↑	GREEN ↑	Brier ↓	ECE ↓
Qwen3.5-9B ^{ZS}	85.55	81.77	13.60	TBD	20.44	31.70	20.22	27.90
LLaVA-v1.6-Mistral-7B ^{ZS}	82.90	73.96	13.67	TBD	19.24	30.78	21.87	29.28
LLaVA-Med-v1.5-7B ^{ZS}	81.59	73.31	1.92	TBD	18.23	21.24	17.61	17.90
CheXagent-8B ^{ZS}	86.89	81.48	0.00	TBD	13.43	15.51	13.00	13.22
BioViL-T + predictor	84.30	81.24	22.65	TBD	14.70	15.37	10.75	7.73
CheXWorld + predictor	83.80	79.56	23.71	TBD	13.15	12.63	11.73	9.04
Copy Current	—	—	0.00	TBD	20.77	24.58	—	—
Finding-transition prior	75.48	64.60	TBD	—	—	—	13.22	7.19
Full-token forecaster (Qwen3.5-9B)	83.09	76.42	16.03	TBD	17.11	17.83	11.64	9.42
MedWorld-JEPA (Qwen3.5-9B slots)	82.70	77.94	28.51	TBD	15.96	17.61	11.47	8.93
Qwen3.5-9B (shuffled state)	70.93	52.85	6.04	TBD	10.78	8.87	13.33	4.97

AP is non-interpolated average precision; AP/AUROC/Brier/ECE are macro-averaged over findings with common reference masks. RadGraph uses partial F1. VLM scores use Yes/No likelihoods. Trained forecasters use 16,000 pairs, 2,400 updates and batch 32; actual GPU hours differ. “+ predictor” is an official frozen-encoder adaptation with shared forecast modules, not an original published forecasting system. The bottom three rows are legacy JEPA-path Qwen3.5-9B forecasters with shared Stage-1 initialization. Their slots use eight final-language-layer queries, distinct from the proposed multi-depth fusion/vision architecture; the full-token control retains the same predictor. Native-image-conditioned 4+4 results, Direction labels and the prior’s transition readout are pending. The prior reads structured source findings. TBD: pending; —: no score for this interface. Single-seed point estimates.

Table 1 results: future prediction

Predict the unobserved follow-up from the current radiograph, current report, source-available clinical history, and requested horizon. Results available on 15 September 2026 are entered below. The public Qwen baseline and the three completed legacy forecast conditions use Qwen3.5-9B. A new native-image-conditioned 4+4 comparison is in progress.

Scoring interface. For each finding, score the fixed candidate answers `Yes` and `No` from the VLM’s conditional token likelihoods:

$$s_k = \frac{\exp(\ell_{\text{Yes},k})}{\exp(\ell_{\text{Yes},k}) + \exp(\ell_{\text{No},k})}.$$

Here ℓ is the candidate answer’s log-likelihood at the same answer boundary. Trained forecasters use finding heads. AP/AUROC assess ranking; Brier/ECE assess the same raw scores as probabilities. These are distinct training regimes.

Evaluation status. The 297-pair reference masks are shared. Transition F1 averages supported finding-specific on-set/resolution events. All entered GREEN scores cover all 297 pairs. Direction labels for this cohort remain unreviewed; scores from the separate 82-pair pilot are excluded. The bottom three rows share a newly trained 9B Stage 1 initialization and have completed matched forecast training and evaluation. Historical 0.8B scores are retained in the experiment records. The 9B slot adaptation uses final-language-layer queries, distinct from the proposed multi-depth fusion/vision model. The new comparison retains native current-image and history inputs at the decoder and adds predicted future slots; its results are not yet available. CheXagent inference and scoring are complete; the prior’s transition readout remains pending. Source paths, hashes and full-precision values are saved in `tables/results_20260914.json`.

Table 2: Available results and remaining evaluations on six downstream tasks. Higher is better. All scores are multiplied by 100 except PSNR (dB); values are shown to two decimals. ZS denotes no benchmark-specific adaptation.

Method	Classification		VQA		Report gen.		Grounding		Segmentation		SR 4×	
	AUC	AP	Acc.	μ F1	RG	CB	mIoU	A@.5	Dice pseudo	Dice human	PSNR	SSIM
Qwen3.5-9B ^{ZS}	71.54	85.46	TBD	TBD	23.98	37.97	TBD	TBD	—	—	—	—
LLaVA-v1.6-Mistral-7B ^{ZS}	54.22	72.64	TBD	TBD	6.87	2.43	TBD	TBD	—	—	—	—
LLaVA-Med-v1.5-7B ^{ZS}	54.70	72.96	TBD	TBD	6.81	15.66	TBD	TBD	—	—	—	—
CheXagent-8B ^{ZS}	79.08	87.92	TBD	TBD	16.93	30.60	TBD	TBD	—	—	—	—
DINOv2 + heads	68.78	84.13	—	—	—	—	—	—	93.03	75.54	32.06	90.86
CheXWorld + heads	75.05	86.18	—	—	—	—	—	—	93.08	75.72	32.06	90.87
MAIRA-2	—	—	—	—	TBD	TBD	TBD	TBD	—	—	—	—
SwinIR (CXR-adapted)	—	—	—	—	—	—	—	—	—	—	37.27	94.51
Image-only + heads	—	—	—	—	—	—	—	—	91.20	75.71	31.87	90.81
Qwen3.5-9B vision + slots [†]	—	—	—	—	—	—	—	—	92.31	75.38	31.87	90.82
Qwen3.5-9B vision + shuffled [†]	—	—	—	—	—	—	—	—	91.18	75.66	31.86	90.79
Qwen3.5-0.8B no-slots	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MedWorld-JEPA (full model)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Classification/report tests contain 353/507 images; classification heads use 13,681 training images and 20 epochs. All filled segmentation/SR rows use 4,096/249/447 train/validation/test images, 20 epochs and batch 8. Human Dice uses 138 external Montgomery lung masks; pseudo Dice uses CXAS. [†]Only the frozen native vision tower is used, with four fixed depth-pooled tokens and trained heads; the 9B language model is not used in these rows. Shuffled tokens come from other patients. SwinIR is fully adapted; its training/evaluation takes 8.31 GPU-h versus 0.16 for image-only SR on the same GPU type, so this is not an equal-GPU-hour comparison. RG/CB: RadGraph/CheXbert F1; A@.5: box accuracy at IoU ≥ 0.5 . TBD: pending; —: not evaluated with this interface. Official VQA, MS-CXR grounding and the full six-task adaptation remain pending; derived QA, anatomy localization and report-assisted prototypes are excluded.

Table 2 results: six downstream tasks

Evaluate current-image understanding, language readout, localization, and reconstruction. The six task families are classification, standard VQA, current report generation, phrase grounding, organ segmentation, and spatial 4× super-resolution.

Data. Filled entries use MIMIC-CXR classification, reports and synthetic SR, CXAS pseudo-labels, and external Montgomery human lung masks. All filled dense-task rows use the 4,096-image training cohort. The 18,708-image runs are reported separately in the experiment records. PSNR/SSIM use the same valid image region.

Remaining evaluations. Official MIMIC-CXR-VQA, MS-CXR phrase grounding and matched six-task adaptation of the full proposed model remain pending. The Qwen3.5-9B native row reports zero-shot classification/reporting; its separate dense rows train heads on four fixed tokens from the frozen vision tower. All filled dense rows share data and updates, but actual GPU hours and pretraining differ. The old report-assisted 2× SR and separate two-epoch VLM adaptation pilots are not scores for the full six-task model.